

MOSFET – P-Channel, 1.8 V Specified, POWERTRENCH®

FDN304PZ

General Description

This P-Channel 1.8 V specified MOSFET uses onsemi's advanced low voltage POWERTRENCH process. It has been optimized for battery power management applications.

Features

- -20 V, -2.4 A $R_{DS(on)} = 52 \text{ m}\Omega @ V_{GS} = -4.5 \text{ V}$
 $R_{DS(on)} = 70 \text{ m}\Omega @ V_{GS} = -2.5 \text{ V}$
 $R_{DS(on)} = 100 \text{ m}\Omega @ V_{GS} = -1.8 \text{ V}$
- Fast Switching Speed
- ESD Protection Diode
- High Performance Trench Technology for Extremely Low $R_{DS(on)}$
- SUPERSOT™ -3 Provides Low $R_{DS(on)}$ and 30% Higher Power Handling Capability than SOT-23 in the Same Footprint
- This is a Pb-Free and Halide Free Device

Applications

- Battery Management
- Load Switch
- Battery Protection

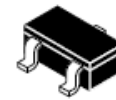
ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Symbol	Parameter	Value	Unit
V_{DS}	Drain-Source Voltage	-20	V
V_{GS}	Gate-Source Voltage	± 8	V
I_D	Drain Current – Continuous (Note 1a) – Pulsed	-2.4 -10	A
P_D	Maximum Power Dissipation (Note 1a) (Note 1b)	0.5 0.46	W
T_J, T_{STG}	Operating and Storage Junction Temperature Range	-55 to +150	$^\circ\text{C}$

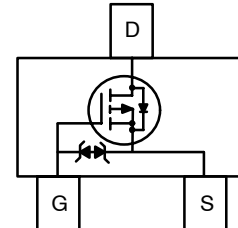
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

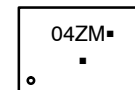
Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient (Note 1a)	250	$^\circ\text{C/W}$
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case (Note 1)	75	$^\circ\text{C/W}$



SOT-23/SUPERSOT™ -23, 3 LEAD,
1.4x2.9
CASE 527AG



MARKING DIAGRAM



04Z = Specific Device Code
M = Month Code
▪ = Pb-Free Package

(Note: Microdot may be in either location)

ORDERING INFORMATION

Device	Package	Shipping†
FDN304PZ	SOT-23 (Pb-Free/ Halide Free)	3000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

FDN304PZ

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

B _V DSS	Drain–Source Breakdown Voltage	V _{GS} = 0 V, I _D = –250 μA	–20	–	–	V
ΔB _V DSS / ΔT _J	Breakdown Voltage Temperature Coefficient	I _D = –250 μA, Referenced to 25°C	–	–13	–	mV/°C
I _{DSS}	Zero Gate Voltage Drain Current	V _{DS} = –16 V, V _{GS} = 0 V	–	–	–1	μA
I _{GSS}	Gate–Body Leakage	V _{GS} = ±8 V, V _{DS} = 0 V	–	–	±10	μA

ON CHARACTERISTICS (Note 2)

V _{GS(th)}	Gate Threshold Voltage	V _{DS} = V _{GS} , I _D = –250 μA	–0.4	–0.8	–1.5	V
ΔV _{GS(th)} / ΔT _J	Gate Threshold Voltage Temperature Coefficient	I _D = –250 μA, Referenced to 25°C	–	3	–	mV/°C
R _{DS(on)}	Static Drain–Source On–Resistance	V _{GS} = –4.5 V, I _D = –2.4 A V _{GS} = –2.5 V, I _D = –2.0 A V _{GS} = –1.8 V, I _D = –1.8 A	–	36 47 65	52 70 100	mΩ
I _{D(on)}	On–State Drain Current	V _{GS} = –4.5 V, V _{DS} = –5 V	–10	–	–	A
g _{FS}	Forward Transconductance	V _{DS} = –5 V, I _D = –1.25 A	–	12	–	S

DYNAMIC CHARACTERISTICS

C _{iss}	Input Capacitance	V _{DS} = –10 V, V _{GS} = 0 V, f = 1.0 MHz	–	1310	–	pF
C _{oss}	Output Capacitance		–	240	–	pF
C _{rss}	Reverse Transfer Capacitance		–	106	–	pF
R _G	Gate Resistance	V _{GS} = 15 mV, f = 1.0 MHz	–	5.6	–	Ω

SWITCHING CHARACTERISTICS (Note 2)

t _{d(on)}	Turn–On Delay Time	V _{DD} = –10 V, I _D = –1 A, V _{GS} = –4.5 V, R _{GEN} = 6 Ω	–	15	27	ns
t _r	Turn–On Rise Time		–	15	27	ns
t _{d(off)}	Turn–Off Delay Time		–	40	64	ns
t _f	Turn–Off Fall Time		–	25	40	ns
Q _g	Total Gate Charge	V _{DS} = –10 V, I _D = –2.4 A, V _{GS} = –4.5 V	–	12	20	nC
Q _{gs}	Gate–Source Charge		–	2	–	nC
Q _{gd}	Gate–Drain Charge		–	2	–	nC

DRAIN–SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS

I _S	Maximum Continuous Drain–Source Diode Forward Current		–	–	–0.42	A
V _{SD}	Drain–Source Diode Forward Voltage	V _{GS} = 0 V, I _S = –0.42 (Note 2)	–	–0.6	–1.2	V
t _{rr}	Reverse Recovery Time	I _F = –2.4 A	–	18	–	ns
Q _{rr}	Reverse Recovery Charge	d _I F/d _t = 100 A/μs	–	7	–	nC

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

NOTES:

- R_{θJA} is the sum of the junction–to–case and case–to–ambient thermal resistance where the case thermal reference is defined as the solder mounting surface of the drain pins. R_{θJC} is guaranteed by design while R_{θCA} is determined by the user's board design.



a) 250°C/W when mounted on a 0.02 in² pad of 2 oz. copper.



b) 270°C/W when mounted on a minimum pad.

Scale 1:1 on letter size paper.

- Pulse Test: Pulse Width ≤ 300 μs, Duty Cycle ≤ 2.0%.

TYPICAL CHARACTERISTICS

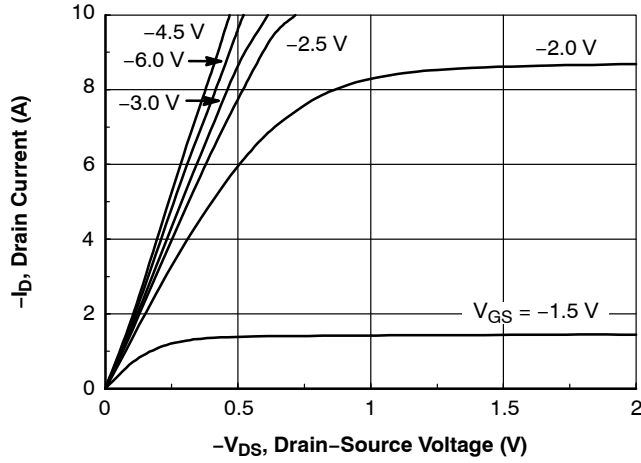


Figure 1. On-Region Characteristics

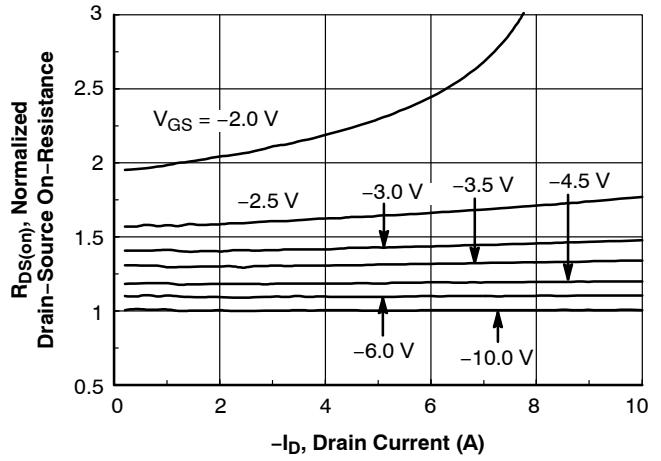


Figure 2. On-Resistance Variation with Drain Current and Gate Voltage

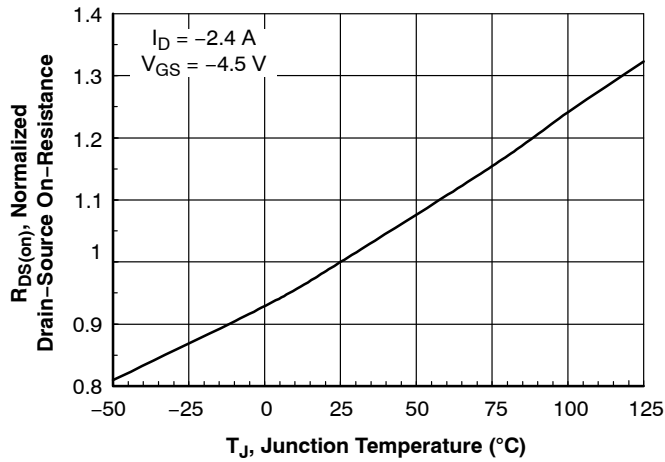


Figure 3. On-Resistance Variation with Temperature

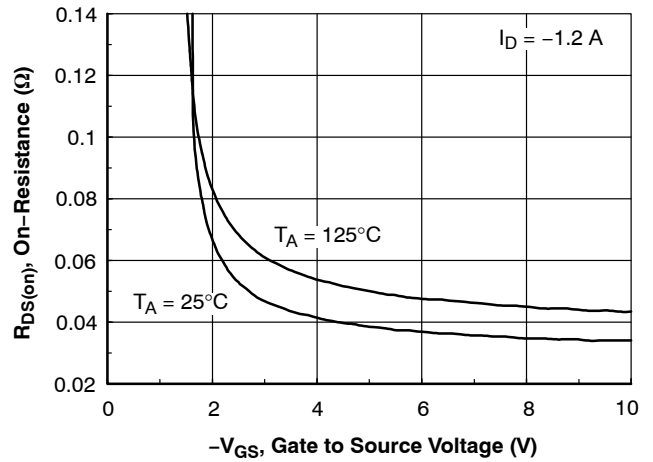


Figure 4. On-Resistance Variation with Gate-to-Source Voltage

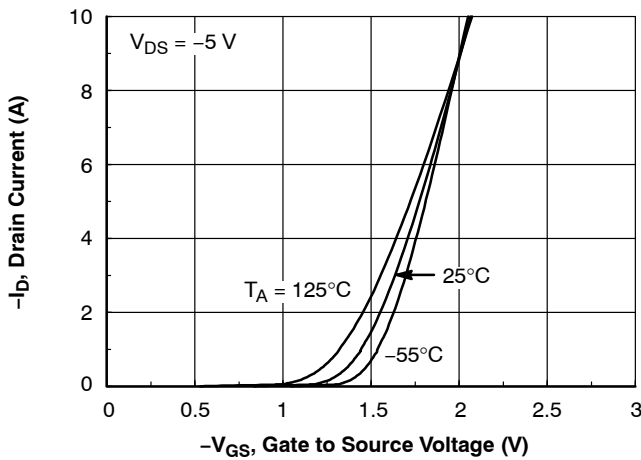


Figure 5. Transfer Characteristics

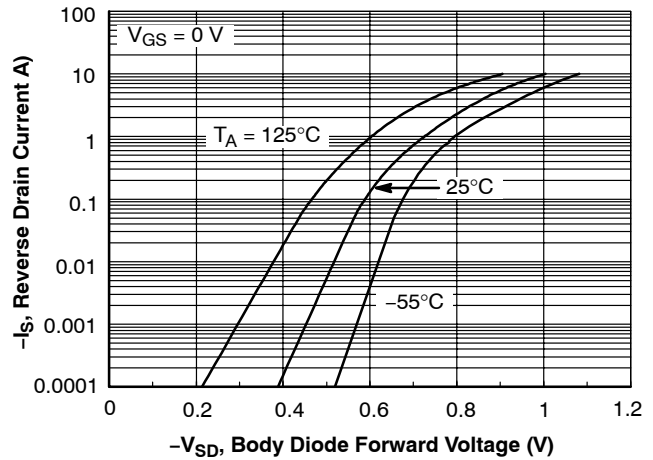


Figure 6. Body Diode Forward Voltage Variation with Source Current and Temperature

TYPICAL CHARACTERISTICS (CONTINUED)

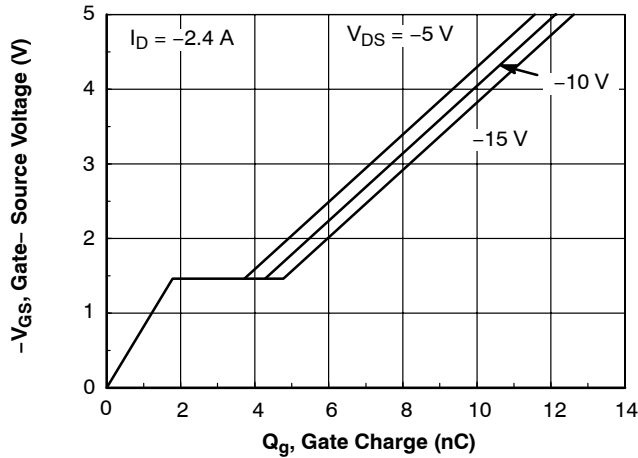


Figure 7. Gate Charge Characteristics

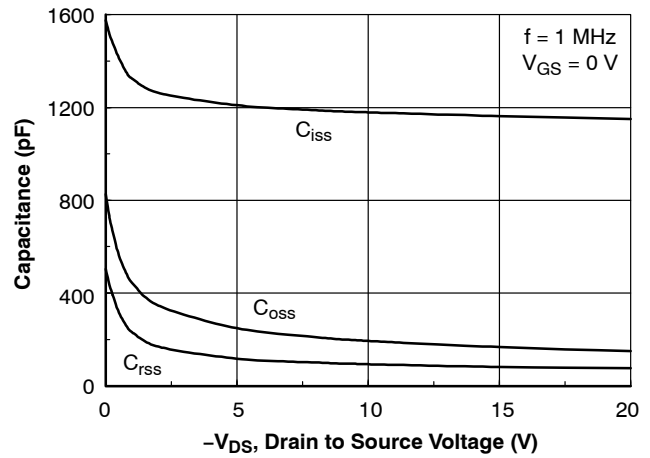


Figure 8. Capacitance Characteristics

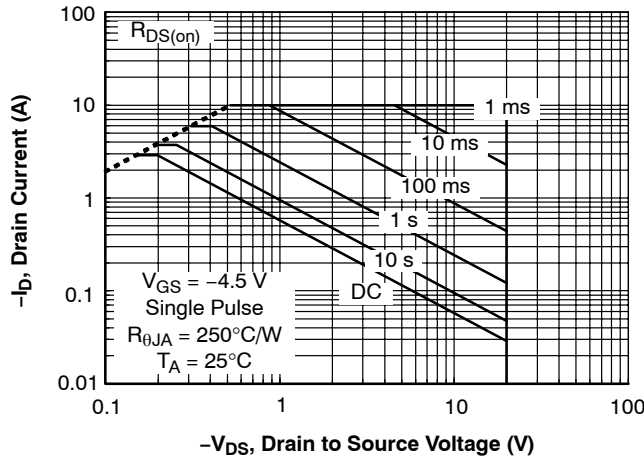


Figure 9. Maximum Safe Operating Area

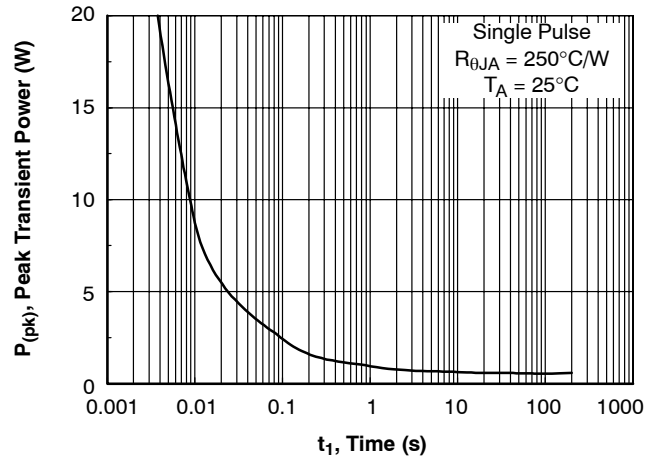


Figure 10. Single Pulse Maximum Power Dissipation

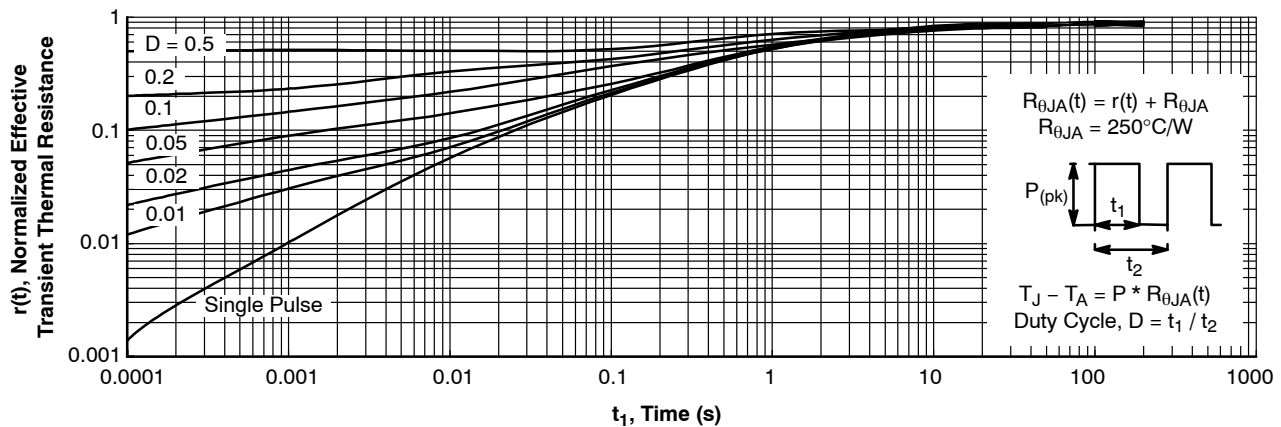


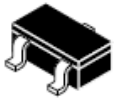
Figure 11. Transient Thermal Response Curve

Thermal characterization performed using the conditions described in Note 1b.

Transient thermal response will change depending on the circuit board design.

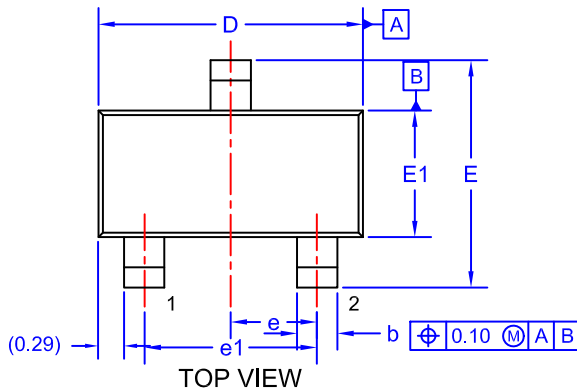
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SOT-23/SUPERSOT™ –23, 3 LEAD, 1.4x2.9
CASE 527AG
ISSUE A

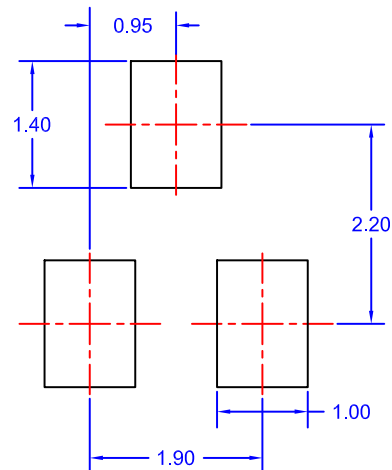
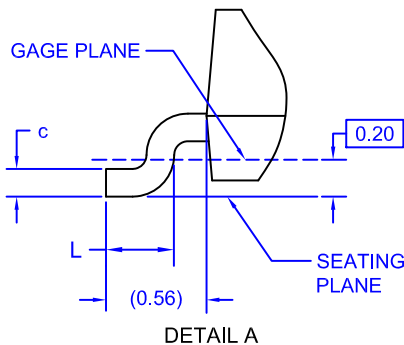
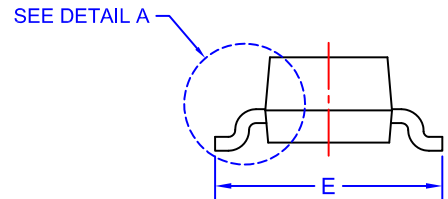
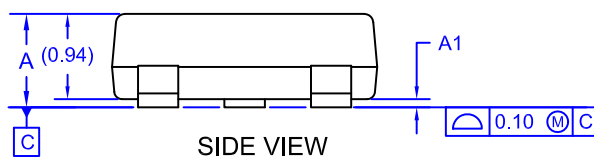
DATE 09 DEC 2019



NOTES: UNLESS OTHERWISE SPECIFIED

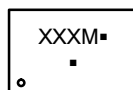
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2. ALL DIMENSIONS ARE IN MILLIMETERS.
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DIM	MIN.	NOM.	MAX.
A	0.85	0.95	1.12
A1	0.00	0.05	0.10
b	0.370	0.435	0.508
c	0.085	0.150	0.180
D	2.80	2.92	3.04
E	2.31	2.51	2.71
E1	1.20	1.40	1.52
e	0.95 BSC		
e1	1.90 BSC		
L	0.33	0.38	0.43



*FOR ADDITIONAL INFORMATION ON OUR Pb-FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
M = Month Code
▪ = Pb-Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

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